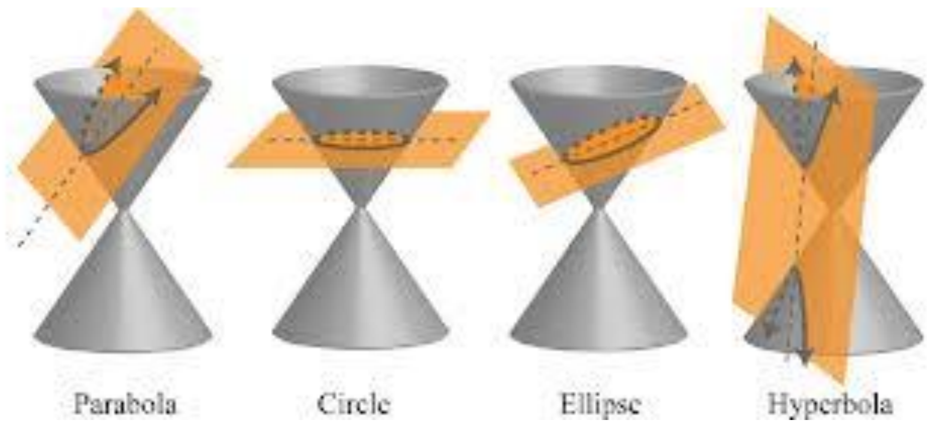
MTH 201 Chapter 5: Parametric/Polar Curves & Conics

5.4.3 Conic Sections (Hyperbolas)

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



Definition: A **hyperbola** is the set of all points in a plane such that the difference between the distances from two fixed points (foci) is constant.

*Note the subtle difference between this definition and the definition for an ellipse which has the *sums* of differences from two fixed points as constant.*

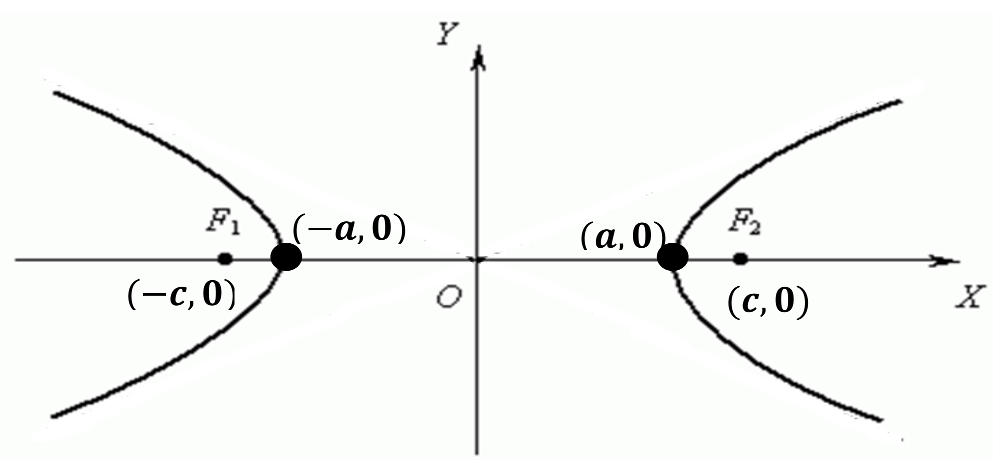
The Equation of a Hyperbola: (opening left or right)

The equation of a hyperbola with center at the origin with foci $(\pm c, 0)$ and x-intercepts $(\pm a, 0)$ is

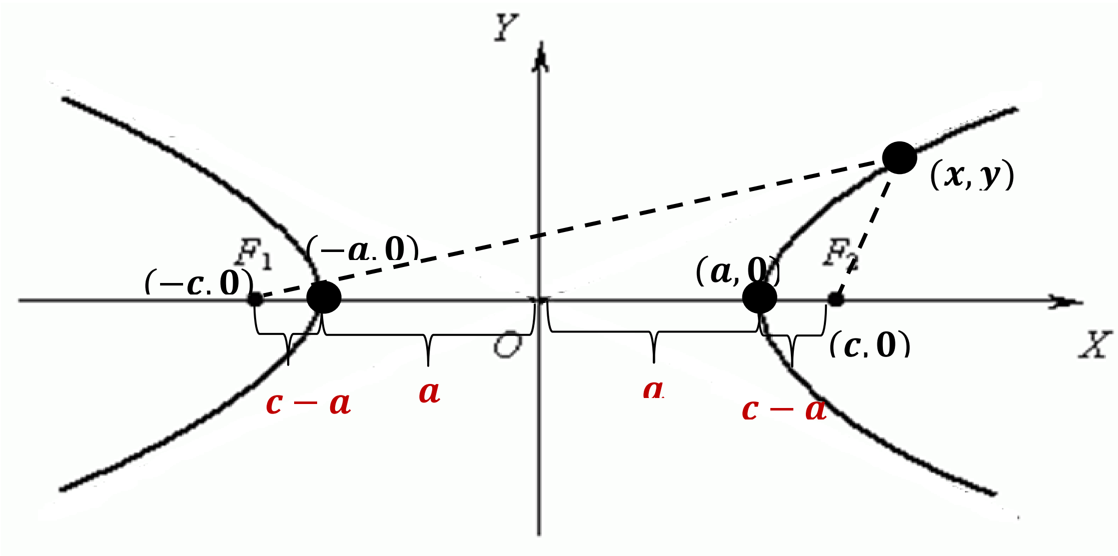
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$

The differences of the distances from any point of the hyperbola to the two foci is *2a*.



Developing the Equation of a Hyperbola *(Not covered in class)*



Note that the distance between $(a,0)$ & $(c,0)$ is $c-a$.

Note that the distance between $(a,0)$ & $(-c,0)$ is $2a+c-a$.

Therefore the constant difference is $2a+c-a-(c-a)=2a$.

So for an arbitrary point (x,y) we want to still have the constant difference *2a*.

$$\sqrt{(x+c)^2+(y-0)^2}+\sqrt{(x-c)^2+(y-0)^2}=2a$$

$$\Leftrightarrow \quad \sqrt{(x+c)^2+y^2}=2a-\sqrt{(x-c)^2+y^2}$$

$$\Leftrightarrow \quad (x+c)^2+y^2=4a^2-4a\sqrt{(x-c)^2+y^2}+(x-c)^2+y^2$$

$$\Leftrightarrow \quad x^2+2cx+c^2+y^2=4a^2-4a\sqrt{(x-c)^2+y^2}+x^2-2cx+c^2+y^2$$

$$\Leftrightarrow \quad 4cx-4a^2=4a\sqrt{(x-c)^2+y^2}$$

$$\Leftrightarrow \quad cx-a^2=a\sqrt{(x-c)^2+y^2}$$

$$\Leftrightarrow \quad c^2x^2-2a^2cx+a^4=a^2\left[(x-c)^2+y^2\right]$$

$$\Leftrightarrow \quad c^2x^2-2a^2cx+a^4=a^2\left[x^2-2cx+c^2+y^2\right]$$

$$\Leftrightarrow \quad c^2x^2-2a^2cx+a^4=a^2x^2-2a^2cx+a^2c^2+a^2y^2$$

$$\Leftrightarrow \quad c^2x^2+a^4=a^2x^2+a^2c^2+a^2y^2$$

$$\Leftrightarrow \quad a^4-a^2c^2=a^2x^2-c^2x^2+a^2y^2$$

$$\Leftrightarrow \quad a^2\left(a^2-c^2\right)=x^2\left(a^2-c^2\right)+a^2y^2$$

$$\Leftrightarrow \quad a^2\left(a^2-c^2\right)=x^2\left(a^2-c^2\right)+a^2y^2$$

Note that since $c>a$ then $c^2>a^2$, so $c^2-a^2>0$. So let $b^2=c^2-a^2$.

$$-a^2\left(c^2-a^2\right)=-x^2\left(c^2-a^2\right)+a^2y^2 \quad \Leftrightarrow \quad -a^2b^2=-x^2b^2+a^2y^2 \quad \Leftrightarrow \quad 1=\frac{x^2}{a^2}-\frac{y^2}{b^2}$$

The Equation of a Hyperbola: (opening up or down)

The equation of a hyperbola with center at the origin with foci $(0,\pm c)$ and x-intercepts $(0,\pm a)$ is

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

where $c^2=a^2+b^2$

The differences of the distances from any point of the hyperbola to the two foci is *2a*.

Note: With hyperbolas it is not important whether a or b is the larger value. We always think of a as the value beneath y and b as the value beneath x.

Graphing a Hyperbola

If we solve the equation $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ for y we obtain

$$\frac{y^2}{b^2}=\frac{x^2}{a^2}-1$$

$$\Leftrightarrow \quad y^2=\frac{x^2b^2}{a^2}-b^2$$

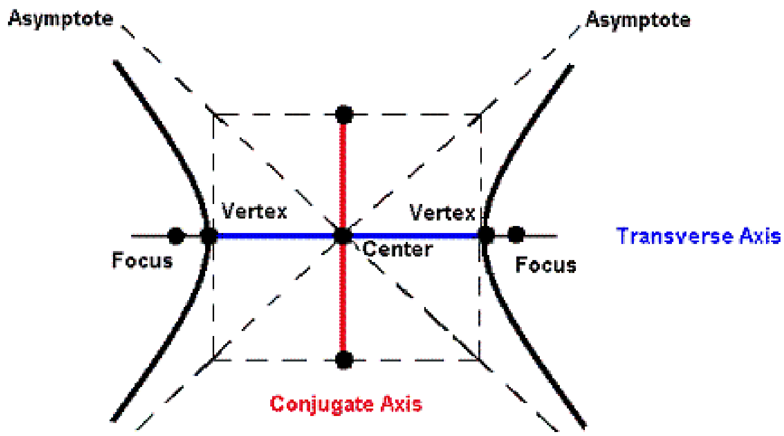
$$\Leftrightarrow \quad y^2=\frac{x^2b^2}{a^2}\left(1-\frac{a^2}{x^2}\right)$$

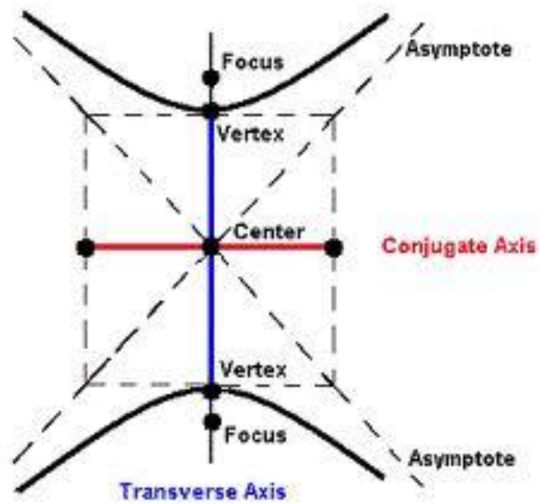
$$\Leftrightarrow \quad y=\pm\frac{bx}{a}\sqrt{1-\frac{a^2}{x^2}}$$

Notice that as $x\to\infty$, we see that $y\to\pm\frac{bx}{a}$

Therefore, we have oblique (slant) asymptotes with the equation: $y=\pm\frac{b}{a}x$

If we have a hyperbola that opens up or down the equation of the asymptotes is given by: $y=\pm\frac{a}{b}x$





Example 1: Determine the foci, equations of the asymptotes, and sketch the graph of

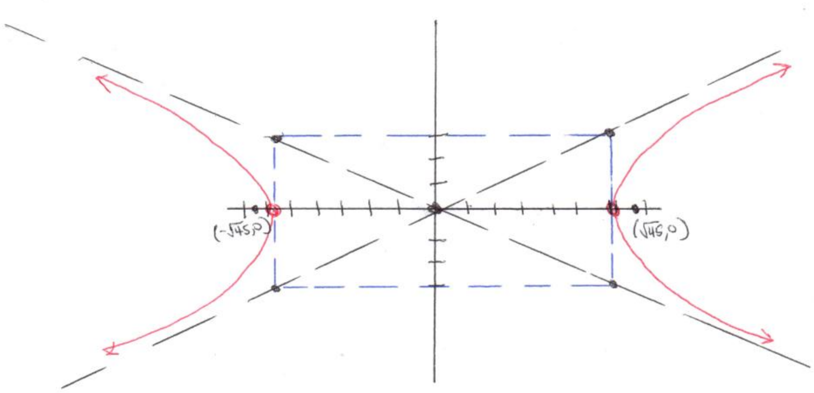
$$\frac{x^2}{36} - \frac{y^2}{9} = 1$$

We have $a^2 = 36,\ b^2 = 9,\ \&\ c^2 = a^2 + b^2 = 36 + 9 = 45$,
So the foci of the hyperbola are

$$\left(\pm\sqrt{45}, 0\right) = \left(\pm 3\sqrt{5}, 0\right)$$

We know that the asymptotes are given by the equation $y = \pm \frac{b}{a} x$

So our asymptotes are: $y = \frac{1}{2}x \quad \& \quad y = -\frac{1}{2}x$



Example 2: Determine the foci, equations of the asymptotes, and sketch the graph of

$$4y^2 - 9x^2 = 36$$

First we divide each side by 36 to get the equation in standard form.

$$\frac{y^2}{9}-\frac{x^2}{4}=1$$

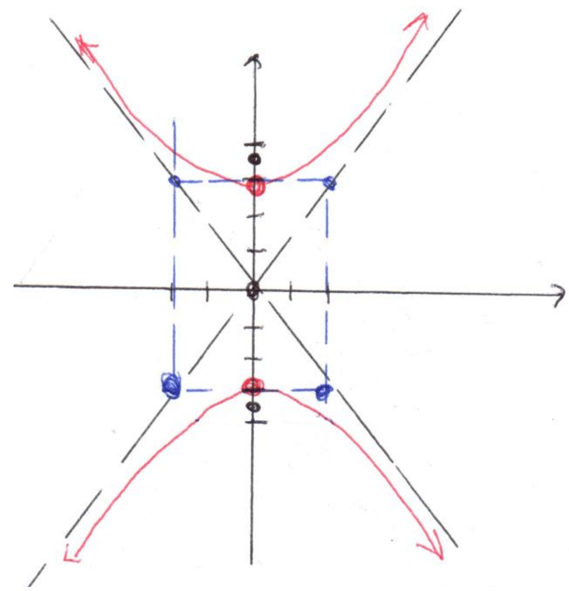
We have $a^2=9$, $b^2=4$, & $c^2=a^2+b^2=9+4=13$

So the foci of the hyperbola are

$$\left(0,\pm\sqrt{13}\right)$$

We know that the asymptotes are given by the equation $y=\pm\frac{a}{b}x$

So our asymptotes are: $y=\frac{3}{2}x$ & $y=-\frac{3}{2}x$



Example 3: Determine the foci, equations of the asymptotes, and sketch the graph of

$$\frac{(x-2)^2}{9}-\frac{(y+1)^2}{4}=1$$

The center of our graph has been shifted from the origin to the point: $(2,-1)$.

We have $a^2=9$, $b^2=4$, & $c^2=a^2+b^2=9+4=13$

So the foci of the hyperbola would have been the points $\left(\pm\sqrt{13},0\right)$ had we still been centered at the origin.

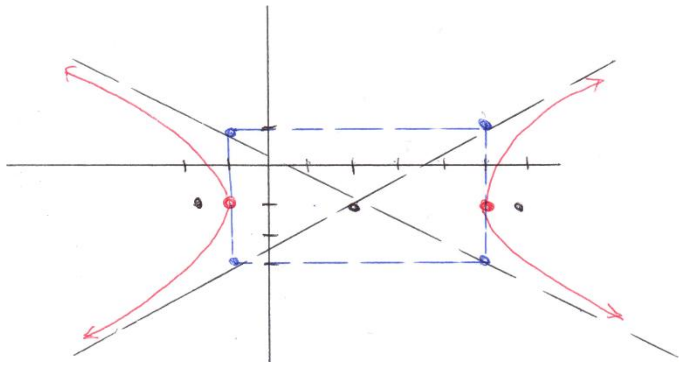
Since we are centered at the point $(2,-1)$ we must add 2 to the *x*-coordinate and add -1 to the *y*-coordinate. So now the foci are

$$\left(2\pm\sqrt{13},-1\right)$$

Similarly we can see that the asymptotes of the graph would have been $y=\pm\frac{2}{3}x$ had the graph been centered at the origin. Since we are centered at the point $(2,-1)$ we must add 2 to the *x*-coordinate and add -1 to the *y*-coordinate. So now the asymptotes are

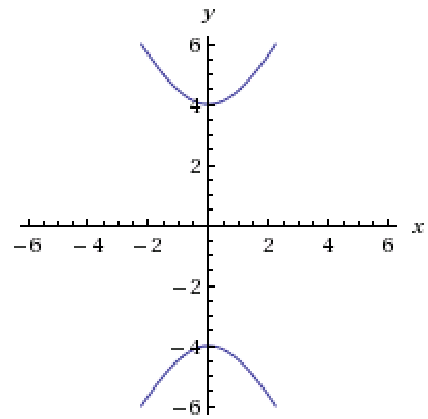
$$(y+1)=\frac{2}{3}(x-2) \quad \& \quad (y+1)=-\frac{2}{3}(x-2)$$

$$\Leftrightarrow \quad y=\frac{2}{3}x-\frac{7}{3} \quad \& \quad y=-\frac{2}{3}x+\frac{1}{3}$$



Example 4: Find the equation of a hyperbola whose transverse axis had endpoints $(0,\pm 4)$ and whose conjugate axis has endpoints $(\pm 2,0)$.

The transverse axis is the distance between the vertices so we see that our hyperbola opens up and down and that $a=4$ which implies that $b=2$.

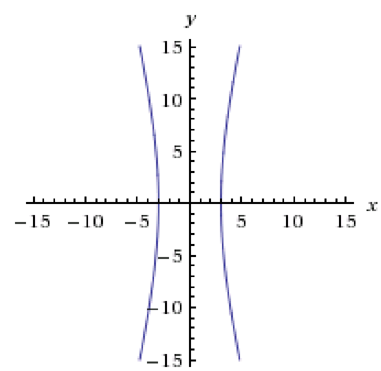


So our equation is:

$$\frac{y^2}{16}-\frac{x^2}{4}=1$$

Example 5: Find the equation of the hyperbola with asymptotes $y=\pm 4x$ and vertices $(\pm 3,0)$.

Since the vertices are $(\pm 3,0)$ we know the hyperbola is centered at the origin and $a=3$. Since the asymptotes go through the vertices of the fundamental rectangle, we know they pass through the point $(3,12)$. So we have that $b=12$.



Therefore, the equation is:

$$\frac{x^2}{9}-\frac{y^2}{144}=1$$

Recapping Section 5.4

Example 6:

(a) Find an equation for the parabola with vertex $(2,3)$, axis of symmetry $x=2$, and which contains the point $(1,5)$.

(b) Find the vertex/vertices of the conic determined by the equation $9x^2-4y^2-72x+8y+176=0$.

(c) Find an equation of the ellipse with foci $(2,-2)$, $(4,-2)$ and vertices $(1,-2)$, $(5,-2)$.

Solutions:

(a)

Based upon the given we know that this parabola opens upward. Since its vertex is at $(2,3)$ we know that the desired equation is of the form $y=\frac{1}{4p}(x-2)^2+3$. We also know that $(1,5)$ must satisfy this equation. Thus, we have $5=\frac{1}{4p}(1-2)^2+3 \quad \Leftrightarrow \quad 2=\frac{1}{4p} \quad \Leftrightarrow \quad p=\frac{1}{8}$.

Thus, the desired equation is $y=2(x-2)^2+3$

(b)

We must complete the square in order to get the equation in a form that we can recognize.

$9x^2-4y^2-72x+8y+176=0 \quad \Leftrightarrow \quad 9x^2-72x-4y^2+8y=-176$

$\Leftrightarrow 9\left(x^2-8x\right)-4\left(y^2-2y\right)=-176 \quad \Leftrightarrow \quad 9\left(x^2-8x+16\right)-4\left(y^2-2y+1\right)=-176+144-4$ From

$\Leftrightarrow 9(x-4)^2-4(y-1)^2=-36 \quad \Leftrightarrow \quad \frac{(y-1)^2}{9}-\frac{(x-4)^2}{4}=1$.

this we recognize that the equation determines a hyperbola with vertical transverse axis and center $(4,1)$. Since $a=3$, we conclude that the vertices are $(4,4)$ and $(4,-2)$.

(c)

We know that the center of the ellipse has to be at the midpoint of the major axis. Thus, the center is $(3,-2)$. Also we see that $a=2$ and $c=1$. Since $c^2=a^2-b^2$, we know that $b^2=3$. Thus, the desired equation is $\frac{(x-3)^2}{4}+\frac{(y+2)^2}{3}=1$.